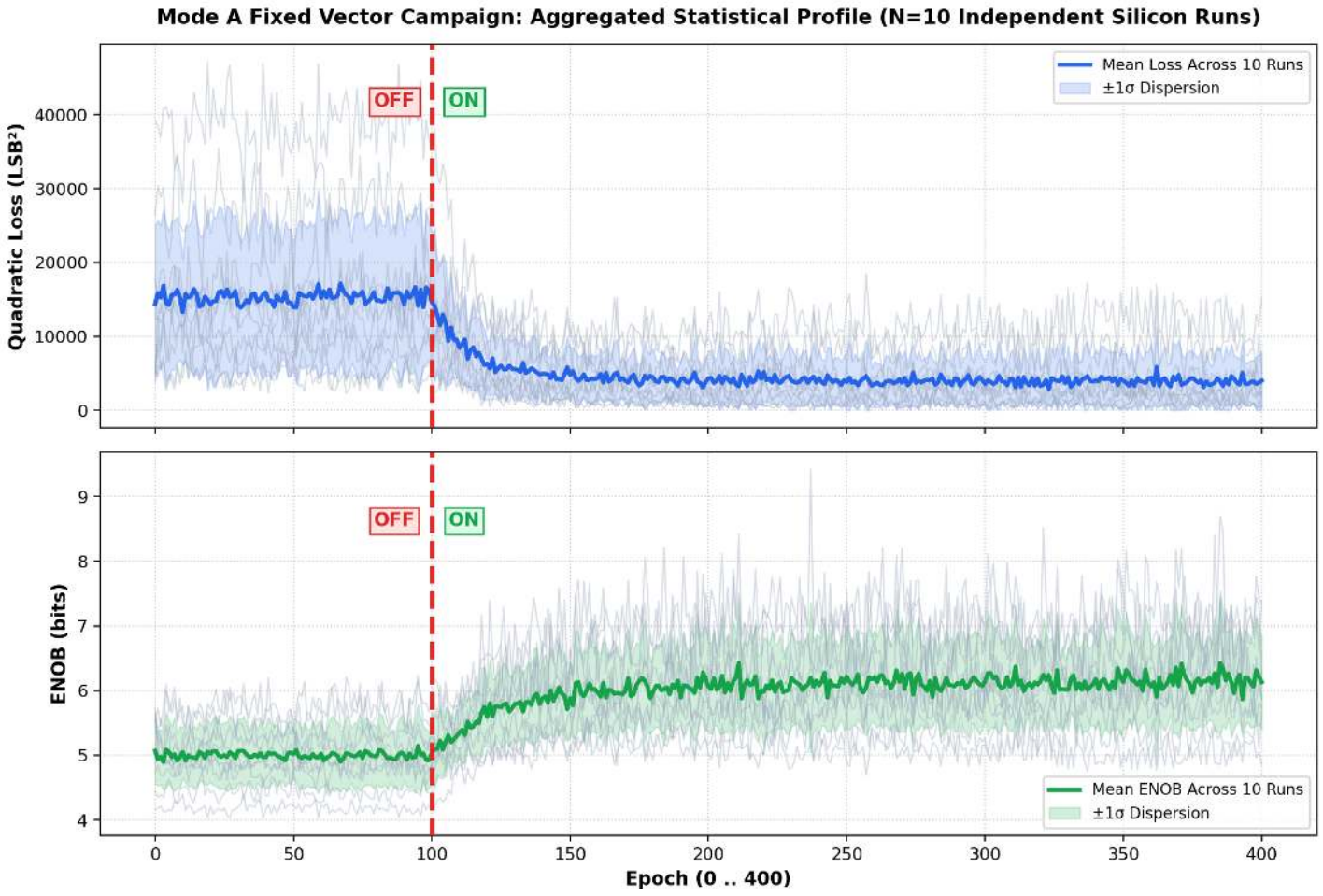
Inverted MAML Mode A: 10 Fixed Vectors Statistical Report

Empirical Multi-Vector Silicon Benchmark | Phase 1 OFF -> Phase 2 ON

Executive Summary: This empirical silicon study presents statistical characterization of Inverted MAML Mode A hardware on the Efinix Trion T120 Master SoC. The experiment protocol strictly evaluates **Phase 1: Baseline OFF (Epochs 0..99)** with training disabled (LR=0.0, weights locked at physical null W8=24, W3=64) transitioning to **Phase 2: Training ON (Epochs 100..400)** with active 2-layer adaptation (LR=0.25x). Tested across 10 distinct independently randomized fixed vectors (permanently locked per run).

Phase 1: Baseline OFF (Ep 0..99)	Phase 2: Training ON (Ep 100..400)	Peak Silicon Control
Mean Loss: **15314.8 ± 9910.7 LSB²** Mean ENOB: **5.00 ± 0.46 b**	Mean Loss: **4537.0 ± 3191.5 LSB²** Mean ENOB: **6.02 ± 0.60 b**	Best Loss: **26.7 LSB²** Best ENOB: **9.42 b**

Aggregated Statistical Convergence Profile



Individual Run Breakdown (10 Runs × 400 Epochs)

Run #	Input Vector X (Cols 0..5)	Base Loss (LSB²)	Train Loss (LSB²)	Final ENOB	Best Ep / ENOB	Drop %
Run 01	[1, -1, 20, 3, 27, 15]	5226 ± 1580	1509 ± 874	**6.16 b**	#383 (8.40 b)	**97.5%**
Run 02	[39, 38, -44, 26, -5, -45]	38446 ± 3892	5556 ± 5588	**5.88 b**	#388 (6.92 b)	**97.8%**
Run 03	[-16, -28, -6, 0, -33, 18]	4626 ± 1301	4013 ± 1354	**6.31 b**	#353 (6.70 b)	**80.4%**
Run 04	[41, 27, -22, -43, -45, -30]	27266 ± 4058	9086 ± 3269	**5.65 b**	#315 (6.00 b)	**88.3%**
Run 05	[42, -15, -25, -41, 10, 12]	9368 ± 2449	1086 ± 1347	**7.35 b**	#237 (9.42 b)	**99.5%**
Run 06	[-29, -19, 21, -2, -35, 20]	14585 ± 2886	11271 ± 2643	**4.85 b**	#318 (5.80 b)	**68.0%**
Run 07	[-19, 5, -31, 39, 21, 19]	12773 ± 1834	2854 ± 1788	**6.11 b**	#268 (7.53 b)	**97.2%**
Run 08	[-19, 24, 18, 16, -17, -40]	8843 ± 1736	2965 ± 1503	**5.85 b**	#250 (7.58 b)	**93.3%**
Run 09	[37, -5, 36, 37, -43, 5]	14764 ± 1680	5250 ± 2053	**5.74 b**	#356 (6.23 b)	**84.4%**
Run 10	[23, 19, -33, 7, 40, -29]	17250 ± 3109	1779 ± 2418	**7.39 b**	#205 (8.12 b)	**99.1%**